

Dr. Renjith Thomas

PhD in Computer Vision

Approved APJAKTU Research Supervisor

Scouting for a progressive career in the area of teaching in a reputed organization by contributing the best of my knowledge and experience to the organization.

PERSONAL DETAILS

- Date of Birth: 05/07/1984
- Sex: Male
- Nationality: Indian
- Religion: Christian
- Marital Status: Married
- Children: 3

CONTACT

Ampattkunnel (H),
Puthusserry (PO),
Mallappally,
Pathanamthitta,
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Research Supervisor

APJAKTU Research Supervisor
3 Scholars doing PhD

WORK EXPERIENCE

Associate Professor & HoD

Department of Artificial Intelligence & Data Science
St Joseph's College of Engineering and Technology Palai
from 19 Jan 2024 to till date

Associate Professor

Musaliar College Pathanamthitta from 03 July 2019 to 18 Jan 2024

Assistant Professor

GISAT Kottayam from 18 June 2012 to 2 July 2019 (7 Years).

Lecturer

Government Polytechnic College Nedumkandam for 8 Months.
Guest Lecturer

Software Engineer

XUSLIMITS Kollam, 3.5 Years. Software Programmer

EDUCATION BACKGROUND

PhD

Hindustan Institute of Science and Technology Chennai

- Image Processing, 30 April 2019
- 1 SCIE indexed Journals
- 2 Scopus Journals
- 2 International IEEE Conferences

MTech

- Anna University Chennai
- Applied Electronics, 2010-2012
- 2 National Conference Papers

BTech

- MGU Kottayam
- Electronics and Communication Engineering, 2002-2006

ORCID: https://orcid.org/0000-0002-1038-9596
Researcher ID: https://publons.com/researcher/4624495/renjith-thomas/
Scopus ID: https://www.scopus.com/authid/detail.uri?authorId=57192674628
Google Scholar: https://scholar.google.com/citations?user=oCnr0xsAAAAJ&hl=en

Google Scholar Citations since 2017: **206 Citations** as on 31/10/2025

Major Financial Assistance

- APJAKTU - CERD - Financial Assistance to Student Projects 2024-25. Principal investigator: Dr Renjith Thomas, Title of the Project: LASER ARMED SURVEILLANCE ENGINE, Total amount sanctioned **Rs 21,000/-**
- KTU Conference grand worth **Rs 34,000/-.**
- Secure a grant of **1 lakh rupees** from the ATAL Academy to coordinate a 6-day ATAL Online FDP.
- GenAI.Works #BuildwithAI International Hackathon, 5th Place with price money of **750$** and one NVDIA Graphis Card.
- Mentor of Prototype Demonstration program organized by the Start up cell of APJAKTU held at College of Engineering Trivandrum on 26-10-2024. Got the grant of **Rs 25000/-**
- Kerala Startup Mission - IDEA Fest 2022 awarded **Rs 2,00,000** (Two Lakhs) as seed money.

Major Activities

- **Invited Talk** in **FDP** on Journal Publication at Mahaguru Institute of Technology, Alapuzha.
- **Invited Talk**: Mandiram Hospital Kottayam on Evolution of AI in the Medical Field on 3/05/2025
- **Keynote speaker** for the FDP organized by St Thomas College Chenganoor.
- **Resource person** of seminar on Revolutionizing Hospitality: Role of AI on 24th January 2025
- Online **Invited Talk** on Introduction to Artificial Intelligence at Eduwing Education Services, Dubai on 2nd February 2025
- **Chief Guest** of Cherubim Fest 2024, Inter School Kids Fest, on Saturday, November 2, 2024
- Presented a paper in IEEE International Conference CINS 2024 held at **BITS Pilani Dubai Campus** on 28 Nov 2024.
- **Reviewer** of Springer Nature Journal 'Aquaculture International'
- **Reviewer** of International Journal of Imaging Systems and Technology by Wiley Publications.
- **Reviewer** for 2nd IEEE International Conference on Networks, Multimedia and Information Technology (NMITCON)
- **Reviewer** of International Conference on Computational Intelligence and Digital Technology ICCIDT 2K24, AIP Publishing
- **Reviewer** of Hindawi Publications 2020.
- International Conference **Editor** (**Scopus Indexed**)- ITechCET-2021 - AIP (American Institute of Physics) 2021.
- International Conference **Editor** (**Scopus Indexed**)- ITechCET-2022 - AIP (American Institute of Physics) 2022.
- **Session Chair** IEEE students project competition Cochin sub session at Carmel Engineering college Alapuzha.
- **Evaluator** of INNOBOR- a 24 hour Hackathon conducted as part of PRAYANA 24 by IEEE RAS Kerala Chapter.
- **Session Chair**, RTACT 2023 National conference.
- **Working demo** of AI based offence detection at Ente Keralam program conducted by Govt of Kerala.
- **Organized** 1 week FEP on NLP and XAI: Faculty Immersion in the era of LLM Sponsered by TCS 2025
- **18 International** IEEE Scopus Indexed Conference Paper Published.
- **3 Patents** and 3 Design of registration.
- 3 **NPTEL MOOC** Courses.

Major Participation

- Successfully participated & completed AICTE Training And Learning (ATAL) Academy Faculty Development Program on Climate Change from 20/01/2025 to 25/01/2025.
- Participant of International Symposium organized by IISR Delhi
- Successfully completed the EBSCO Training on EBSCO eBooks/ECM
- Participated in the online training/webinar on "IEEE Xplore"
- Participated one day regional meeting organized by Additional Skill Acquisition Programme, Kerala on 27th may 2024

Official Assignments

- University Question Paper Setter of 4 Courses 2024-2025 & 2025-2026
- Syllabus Revision of KTU 2024, PEADT 526 Computational Biology and MTech Syllabus
- SJCET college council member, Anti-ragging committee member, IQAC faculty representative, academic audit member, research advisory board member, Academic council member
- Coordinator of Ente keralam program from institution-Winner Open Area Stall 2023.
- NBA Program Coordinator (Got NBA in ECE 2023-2026)
- Dean Research 2021-2023.

RESEARCH PUBLICATIONS

Journals

1. J. A. Jose, Renjith Thomas, (2026). Zero shot multistage hierarchical fish classification using YOLOE and large language model, Computers and Electronics in Agriculture, Volume 242, 2026, 111288, ISSN 0168-1699,
2. Thomas, R., & Rangachar, M. J. S. (2017). Fractional bat and multi-kernel-based spherical SVM for low resolution face recognition. International Journal of Pattern Recognition and Artificial Intelligence, *World Scientific*, 31(08), 1756014.
3. Thomas, R., & Rangachar, M. J. S. (2016). Integrating GWTM and BAT algorithm for face recognition in low-resolution images. *The Imaging Science Journal*, 64(8), 441-452.

Patent

1. **Title of invention: Secure Device To Device Connectivity Mechanism for IoT Smart City Applications.**

Publication date: 28/04/2021
Patent office : **IP Australia**
Patent 202641031698 A 2021101443

2. **Title of invention: AI Enabled Under Water Pollution Tracking Device.**

Design number: 6336195
Publication date: December 2023
Patent office : **UK Design Patent**
Grant date: 13 January 2024

3. **Title of invention: Smart HealthCare Monitoring Apparatus**

Application Number: 424211-001
Cbr Number: 212897
Cbr Date: 22/07/2024 19:41:17
Patent office : Certificate of Registration of Design, **IP India.**

4. **Title of invention: Integrated PV Solar Cell Enclosure System Providing Corrosion and Moisture Protection for Padlocks and Similar Devices**

Publication date: 12/12/2025
Patent office : **IP India**
Application No: 202541116009

5. **Title of invention: Assistive Writing Instrument Incorporating Ball-Bearing Support For Reduced-Grip Users**

Publication date: 20/02/2026
Patent office : **IP India**
Application No: 2021101443

6. **Title of invention: Calibrated liquid dispensing apparatus with screw actuated plunger mechanism**

Publication date: 16/03/2026
Patent office: IP India
Application No: 202641031698 A

Patent cont'd

7. Title of invention: Hybrid Physical-Digital Authentication System using Structured Surface Geometry and AI-Based Verification

Publication date: 24/04/2026
Patent office: IP India
Application No: 202641046647 A

8. Title of invention: Wearable Multi-Node Thermoelectric and Haptic Therapy Device for Simultaneous Multi-Modal Thermal Stimulation

Publication date: 01/05/2026
Patent office: IP India
Application No: 202641049237 A

9. Title of invention: Pedal-Driven Coconut Processing Apparatus with Flywheel-Assisted Torque Transfer and Integrated Deshelling and Scrubbing Mechanism

Publication date: 12/06/2026
Patent office: IP India
Application No: 202641069944 A

10. Title of invention: Grip-Pressure Actuated Retractable Writing Instrument Incorporating Linkage and Rack-and-Pinion Mechanism

Publication date: 26/06/2026
Patent office: IP India
Application No: 202641074284 A

11. Title of invention: Foot-Operated Surface Cleaning Apparatus with Brush Actuation and Liquid Dispensing Mechanism

Publication date: 202641079225 A
Patent office: IP India
Application No: 03/07/2026

12. Title of invention: Automated Sequential Multi-Chamber Infant Urine Collection Device for Sterile Midstream Specimen Isolation

Publication date: 03/07/2026
Patent office: IP India
Application No: 202641080228 A

13. Title of invention: Integrated Endoscopic Hydro-Jetting System for Simultaneous Inspection, Cleaning, and Sanitary Pipeline Remediation

Publication date: 17/07/2026
Patent office: IP India
Application No: 202641086056 A

14. Title of invention: Dual-stroke, variable-ratio, self-locking mechanical chain lifting system for bidirectional lever actuation

Publication date: 24/07/2026
Patent office: IP India
Application No: 202641088572 A

15. Localized induction heating-based ironing system with selective coil activation and passive ironing handle

Publication date: 31/07/2026
Patent office: IP India
Application No: 202641090848

Conferences

1. A. Varghese, J. Joseph and R. Thomas, "An Automated Platform for Habituation Analysis in Mimosa Pudica Using an Automated Habituation Assessment (AHA) Algorithm," 2026 International Conference on Intelligent and Innovative Technologies in Computing, Electrical and Electronics (IITCEE), Bangalore, India, 2026, pp. 1-6, doi: 10.1109/IITCEE67948.2026.11394635.
2. A. M. Varghese, J. Biju, N. J. Manasrayil, J. Philip, M. S. Saritha and R. Thomas, "A Feedback-Driven MLOps Based Approach for Fall Detection," 2025 IEEE International Conference on Contemporary Computing and Communications (InC4), Bangalore, India, 2025, pp. 1-7, doi: 10.1109/InC465408.2025.11256398.
3. A. Sreenivasan, R. Antony, G. Joseph, S. Sunny, N. Tom and R. Thomas, "Automated E-Waste Segregation and Valuation using AI and Image Recognition," 2025 7th International Conference on Innovative Data Communication Technologies and Application (ICIDCA), Coimbatore, India, 2025, pp. 908-913, doi: 10.1109/ICIDCA66325.2025.11280439.
4. A. R. Benny, J. George, J. Johny, J. J. Raju, A. James and R. Thomas, "Automated Trading Bot using Long Short-Term Memory," 2025 Annual International Conference on Data Science, Machine Learning and Blockchain Technology (AICDMB), MYSURU, India, 2025, pp. 1-6, doi: 10.1109/AICDMB64359.2025.11277509.
5. A. Satheish, S. Jimmy, Sanjana, S. Thomas and R. Thomas, "Foot Length Measuring System Using OpenCV and ResNet50," 2025 International Conference on Intelligent Communication Networks and Computational Techniques (ICICNCT), Bidar, India, 2025, pp. 1-6, doi: 10.1109/ICICNCT66124.2025.11232598.
6. A. George, E. J. Muttom, J. Joyan, N. Binoy, A. James and R. Thomas, "ParkIt: An ANPR based Automated Parking Registry Using YOLO And EasyOCR," 2025 2nd International Conference on Trends in Engineering Systems and Technologies (ICTEST), Ernakulam, India, 2025, pp. 1-6, doi: 10.1109/ICTEST64710.2025.11042700.
7. A. George, E. J. Muttom, J. Joyan, N. Binoy, J. Shogi and R. Thomas, "A Personalized AI Assistant for Analyzing and Authoring Product Reviews using Groq/Llama3-70B," 2025 International Conference on Computing Technologies & Data Communication (ICCTDC), HASSAN, India, 2025, pp. 1-6, doi: 10.1109/ICCTDC64446.2025.11158767.
8. A. R. Benny, J. George, J. Johny, J. J. Raju, A. James and R. Thomas, "Offline Proctoring Using YOLOv8 and VGG-16," 2025 Annual International Conference on Data Science, Machine Learning and Blockchain Technology (AICDMB), MYSURU, India, 2025, pp. 1-6, doi: 10.1109/AICDMB64359.2025.11277919.
9. B. Paul, A. Nair, A. S, A. Thomas, N. Tom and R. Thomas, "Geo-Location Based Job Portal," 2025 International Conference on Advancements in Power, Communication and Intelligent Systems (APCI), Kannur, India, 2025, pp. 1-6, doi: 10.1109/APCI65531.2025.11136893.
10. A. M. Shibu, A. Tom, R. M. Mathew, R. B. Mathews, J. Thomas and R. Thomas, "AI-Enhanced Emergency Call Center Assistant and Situational Awareness for Optimized Emergency Response: Using Speech Recognition, NLP Mapping, NER," 2025 4th International Conference on Advances in Computing, Communication, Embedded and Secure Systems (ACCESS), Ernakulam, India, 2025, pp. 85-90, doi: 10.1109/ACCESS65134.2025.11135720
11. Varghese, A.M., Biju, J., Manasrayil, N.J., Shaji, L. and Thomas, R., 2025, February. Human Exclusive Alerts in CCTV Surveillance Using Computer Vision. In 2025 Emerging Technologies for Intelligent Systems (ETIS) (pp. 1-6). IEEE.
12. Sasi, D., Arjun, P., Jinso, J., Paul, J. and Thomas, R., 2025, February. ResoBlend: A Hybrid Approach for Image Quality Enhancement. In 2025 Emerging Technologies for Intelligent Systems (ETIS) (pp. 1-6). IEEE.
13. Sabu, D., Jose, A., Philip, J., Pranav, P., Thomas, R., & Mathew, J. (2024, November). Automated Mark Entry in Educational Institutions Using Multiple CNNs: A Case Study on Recognizing Handwritten Fractional Marks. In 2024 International Conference on Computational Intelligence and Network Systems (CINS) (pp. 1-8). IEEE.
14. R. S. Sen, M. Amalkrishna, R. Prithviraj, S. P. Jose, N. Joseph and R. Thomas, "Simple Action Model: Enabling LLM to Sequential Function Calling Tool Chain," 2024 International Conference on Advancement in Renewable Energy and Intelligent Systems (AREIS), Thrissur, India, 2024, pp. 1-6. IEEE
15. R. Thomas, A. F, A. S. John and S. K. S, "Automated Detection of Traffic Rule Violation using Deep Learning Techniques," 2024 IEEE Recent Advances in Intelligent Computational Systems (RAICS), Kothamangalam, Kerala, India, 2024, pp. 1-6. IEEE
16. Habeeb, N., Thomas, R. and Oommen, K., 2023, December. Automated Detection of Pneumonia Using Pre-Trained Convolutional Neural Networks in X-Ray Images. In 2023 3rd International Conference on Mobile Networks and Wireless Communications (ICMNWC) (pp. 1-6). IEEE.
17. Thomas, R. “Face recognition from image patches using an ensemble of CNN-local mesh pattern networks”. IEEE 6th International Conference for Convergence in Technology (I2CT) (pp. 1-5) 2021.
18. Thomas, R., & Rangachar, M. J. S. GWTM and crow search algorithm for face recognition in low-resolution images”. IEEE International Conference on Advances in Computing, Communication Control and Networking (ICACCCN) (pp. 963-970) 2018.

Books

- Abstract Proceedings of the International Conference on Research Advances in Engineering and Technology - ITechCET 2021, Dr Shan M Assis, Dr Kurian Antony, Dr I Bushra, Dr Ajesh F, **Dr Renjith Thomas,** Dr Teena Joseph, Wizcraft Publications and Distribution.

Publication date: June 2023
ISBN: 9780735443846
https://printorders.aip.org/proceedings/2520

- Abstract Proceedings of the International Conference on Research Advances in Engineering and Technology - ITechCET 2022, Dr Shan M Assis, Dr Ajesh F, Dr I Bushra, Dr D Chithra Prasad, **Dr Renjith Thomas**, Dr Ciby Jacob Cherian, Dr Teena Joseph, Dr Sonia Katherine, Wizcraft Publications and Distribution.

Issue Date: 09/03/2023
ISBN: 978-93-92268-31-1
https://isbnnew.inflibnet.ac.in/Recently_Published_Books.aspx

Achievements

- Shortlisted for **AICTE RPS** research Grand (34 Lakhs)
- Secured fifth position for **#BuildwithAI Hackathon 2024** among 200 teams from across 80 countries.
- Kerala Startup mission - IDEA Fest 2022 awarded **Two Lakhs** as seed money from 17 Lakhs for Artificial Intelligence based vehicle modification detector.
- Secure a grant of **1 lakh** rupees from the ATAL Academy to coordinate a 6-day ATAL Online FDP.
- Organized **1 week FEP** on NLP and XAI: Faculty Immersion in the era of LLM Sponsered by TCS 2025
- Reboot Kerala Hackathon **Finalist** 2019.
- Generated **29 SCI/ Scopus Publications** from the department in 2 years

Professional Membership

IEEE Member #100870822, Valid through 31 December 2026

Co-Founder

DUPICO, DesignDirect – Web & AI Development Company, Enfield, London
(Aug 2025 – Present)

Declaration

I hereby declare that all the above furnished information is true to best of my knowledge and belief.



Date: 29/July/2026 Dr. Renjith Thomas